

Omada OCNA Practice Exam

Omada Certified Network Associate - 60 multiple-choice questions

Instructions: Choose the single best answer for each question. Suggested time: 60 minutes. A passing score of about 70% is a sensible target. The answer key with short explanations is on the last pages.

Note: These are original practice questions written around the published OCNA topic areas (networking fundamentals, switching, wireless, Omada SDN, gateway and security). They are not copies of real exam questions. Port numbers, defaults and menu names can change between controller versions, so verify them against the official TP-Link Omada documentation and the current OCNA exam outline.

Section 1. Networking Fundamentals

1. Which OSI layer is responsible for routing packets between different networks?

- A) Session layer (Layer 5)
- B) Data Link layer (Layer 2)
- C) Network layer (Layer 3)
- D) Transport layer (Layer 4)

2. A standard Layer 2 switch makes forwarding decisions based on which address?

- A) Destination IP address
- B) Source port number
- C) Hostname
- D) Destination MAC address

3. How many usable host addresses are available in a /26 subnet?

- A) 64
- B) 62
- C) 126
- D) 30

4. Which of the following is a private IPv4 address?

- A) 11.0.0.1
- B) 192.169.1.1
- C) 172.20.5.10
- D) 172.32.0.1

5. What is the correct order of the DHCP process?

- A) Offer, Discover, Acknowledge, Request
- B) Discover, Offer, Request, Acknowledge
- C) Request, Discover, Acknowledge, Offer
- D) Discover, Request, Offer, Acknowledge

6. Which protocol resolves an IP address to a MAC address on a local network?

- A) ICMP
- B) DHCP
- C) ARP
- D) DNS

7. Which transport protocol is connectionless and does not guarantee delivery?

- A) TCP
- B) TLS
- C) SCTP
- D) UDP

8. What is the primary purpose of NAT?

- A) Encrypt traffic between sites
- B) Translate private IP addresses to public IP addresses
- C) Resolve domain names
- D) Assign IP addresses automatically

9. What is the network address of host 192.168.10.77/26?

- A) 192.168.10.0
- B) 192.168.10.76
- C) 192.168.10.64
- D) 192.168.10.128

10. Which protocol does the ping command use?

- A) ARP
- B) UDP port 53
- C) SNMP
- D) ICMP

11. Which well-known port is used by DNS?

- A) 443
- B) 53
- C) 67
- D) 80

12. What does the default gateway do for a host?

- A) Blocks all inbound traffic
- B) Assigns IP addresses to clients
- C) Translates domain names
- D) Forwards traffic destined for other networks

Section 2. Switching, VLAN and Link Technologies

13. Which IEEE standard defines VLAN tagging?

- A) IEEE 802.11ac
- B) IEEE 802.3ad
- C) IEEE 802.1Q
- D) IEEE 802.1X

14. What is the valid range of usable VLAN IDs?

- A) 0 to 4095
- B) 1 to 255
- C) 1 to 4094
- D) 1 to 1024

15. Which port type carries traffic for multiple VLANs between switches?

- A) Access port
- B) Isolated port
- C) Mirror port
- D) Trunk port

16. What is the main purpose of Spanning Tree Protocol (STP)?

- A) Prevent Layer 2 loops
- B) Increase port bandwidth
- C) Provide PoE to devices
- D) Encrypt Layer 2 frames

17. Which standard defines Rapid Spanning Tree Protocol (RSTP)?

- A) IEEE 802.1w
- B) IEEE 802.3af
- C) IEEE 802.1D
- D) IEEE 802.1Q

18. Which feature combines multiple physical links into one logical link for more bandwidth and redundancy?

- A) IGMP snooping
- B) Link Aggregation (LACP, 802.3ad)
- C) Port mirroring
- D) VLAN tagging

19. Which PoE standard delivers up to 30 W per port at the power sourcing equipment?

- A) IEEE 802.1p
- B) IEEE 802.3at (PoE+)
- C) IEEE 802.3u
- D) IEEE 802.3af (PoE)

20. What does port mirroring do?

- A) Blocks traffic on unused ports
- B) Duplicates the MAC table to a second switch
- C) Doubles the bandwidth of a port
- D) Copies traffic from one port to another for analysis

21. What is the benefit of IGMP snooping?

- A) Limits multicast traffic to ports that requested it
- B) Assigns IPs to multicast groups
- C) Prevents VLAN hopping
- D) Detects cable faults

22. Which feature limits excessive broadcast, multicast or unknown unicast traffic on a port?

- A) Port isolation
- B) Storm Control
- C) DHCP snooping
- D) Link backup

23. What is the native VLAN on a trunk port?

- A) The VLAN with the highest ID
- B) The VLAN used only for management
- C) The VLAN that is always blocked
- D) The VLAN whose traffic is sent untagged

24. Which feature detects when a cable creates a loop from a switch port back into the same switch?

- A) QoS
- B) Loopback detection
- C) Rate limiting
- D) Port mirroring

Section 3. Wireless Technologies

25. Which three channels are non-overlapping in the 2.4 GHz band with 20 MHz width?

- A) 2, 7 and 12
- B) 1, 6 and 11
- C) 1, 5 and 9
- D) 3, 6 and 9

26. Which IEEE standard is marketed as Wi-Fi 6?

- A) 802.11ad
- B) 802.11ac
- C) 802.11n
- D) 802.11ax

27. Which frequency band was added by Wi-Fi 6E?

- A) 6 GHz
- B) 900 MHz
- C) 60 GHz
- D) 3.6 GHz

28. Which technology lets an AP serve multiple clients within a single transmission by dividing the channel into resource units?

- A) OFDMA
- B) CSMA/CD
- C) STP
- D) Band steering

29. Which Wi-Fi 5 (802.11ac) band is used?

- A) 6 GHz only
- B) 2.4 GHz only
- C) Both 2.4 GHz and 5 GHz exclusively
- D) 5 GHz only

30. Which statement about 2.4 GHz compared to 5 GHz is correct?

- A) 2.4 GHz has more non-overlapping channels
- B) 2.4 GHz has better range and wall penetration but is more congested
- C) 2.4 GHz offers higher maximum throughput
- D) 2.4 GHz is not affected by microwave ovens

31. What does WPA3-Personal use to replace the PSK four-way handshake weakness against offline dictionary attacks?

- A) MAC filtering
- B) WEP
- C) TKIP
- D) SAE (Simultaneous Authentication of Equals)

32. What is the purpose of band steering?

- A) Encourage dual-band clients to connect to the 5 GHz band
- B) Change the SSID automatically
- C) Move clients to the 2.4 GHz band
- D) Encrypt traffic between APs

33. Which technology allows an AP to transmit to multiple clients simultaneously using spatial streams?

- A) Channel bonding
- B) SU-MIMO
- C) MU-MIMO
- D) Beamforming only

34. Which 802.11 amendment defines Fast BSS Transition (fast roaming)?

- A) 802.11e
- B) 802.11w
- C) 802.11r
- D) 802.11s

35. What is the effect of using a wider channel width (e.g. 80 MHz instead of 20 MHz)?

- A) Longer range and fewer interference problems
- B) Lower throughput but better security
- C) Higher potential throughput but fewer non-overlapping channels and more interference
- D) No effect on throughput

36. In a wireless mesh deployment, what is a wireless uplink?

- A) A connection between an AP and a wired switch
- B) A PoE injector connected to an AP
- C) An AP connecting to another AP over the air instead of Ethernet
- D) A client connected to an AP via WPS

Section 4. Omada SDN Solution and Controller

37. Which components form the Omada SDN solution?

- A) Omada Controller, EAP access points, switches and gateways
- B) VLAN-aware routers only
- C) Only EAP access points
- D) Cloud storage and EAP only

38. Which of these is NOT a type of Omada Controller?

- A) Software Controller
- B) Hardware Controller (e.g. OC200)
- C) Cloud-Based Controller
- D) Mobile-only Controller

39. Which TCP port is used by default for HTTPS management of the Omada Software Controller?

- A) 8043
- B) 8088
- C) 443 only
- D) 29810

40. Which UDP port does the Omada Controller use for device discovery?

- A) 29810
- B) 8088
- C) 67
- D) 29811

41. What is a Site in the Omada Controller?

- A) A logical group of devices with its own settings, for example one branch or location
- B) A VLAN
- C) A single wireless SSID
- D) A firmware version

42. What does adoption mean in Omada?

- A) The firmware is downgraded
- B) A client is added to the guest list
- C) The controller takes over management of a device and applies configuration to it
- D) A device is restored to factory defaults

43. What is a key characteristic of Standalone mode for an EAP?

- A) The AP must use Mesh
- B) The AP cannot broadcast an SSID
- C) The AP is managed individually through its own web interface
- D) The AP is managed from a central cloud only

44. What is the benefit of Omada Cloud-Based Controller?

- A) It only works on a single LAN
- B) It replaces the gateway
- C) Centrally manage sites remotely without deploying local controller hardware
- D) It removes the need for access points

45. Which hardware controller models belong to the Omada family?

- A) OC200 and OC300
- B) TL-SG3428 and TL-SG2008
- C) EAP225 and EAP610
- D) ER605 and ER7206

46. Which Omada guest portal type lets the administrator generate time- or data-limited codes?

- A) No Authentication
- B) Simple Password
- C) MAC filtering
- D) Voucher

47. What is the role of a captive portal?

- A) Redirect new clients to an authentication or splash page before granting internet access
- B) Encrypt wireless traffic with WPA3
- C) Perform firmware upgrades
- D) Balance WAN links

48. Which technology provides port-based network access control using a RADIUS server?

- A) 802.11r
- B) 802.1X
- C) 802.3ad
- D) 802.1Q

Section 5. Gateway, Routing, VPN and Security

49. Which feature lets an Omada Gateway use several WAN connections at the same time?

- A) WAN load balancing
- B) Port mirroring
- C) STP
- D) Band steering

50. In Link Backup mode, when is the backup WAN used?

- A) When the primary WAN link fails
- B) Only for guest traffic
- C) Always, at the same time as the primary
- D) Only at night

51. What does policy routing allow?

- A) Creating PoE schedules
- B) Assigning VLAN IDs automatically
- C) Blocking all unknown MAC addresses
- D) Sending specific traffic over a chosen WAN or next hop based on defined rules

52. Which VPN type connects two branch offices permanently?

- A) Client-to-Site VPN
- B) Guest VPN
- C) Captive portal VPN
- D) Site-to-Site VPN

53. Which protocol suite is commonly used to build encrypted site-to-site VPN tunnels?

- A) DHCP
- B) IPsec
- C) ICMP
- D) ARP

54. What is the purpose of an ACL on an Omada Gateway?

- A) Provide PoE
- B) Update device firmware
- C) Increase wireless range
- D) Permit or deny traffic between networks based on defined rules

55. What is a DMZ typically used for?

- A) Increasing WiFi speed
- B) Bundling ports
- C) Assigning guest vouchers
- D) Hosting public-facing servers separated from the internal LAN

56. What does port forwarding (virtual server) do?

- A) Bundles two WAN ports
- B) Disables NAT
- C) Copies traffic to a monitoring port
- D) Maps an external port on the WAN to an internal host and port

57. What is the advantage of using VLANs for segmentation, such as separate Guest and IoT networks?

- A) Removes the need for a gateway
- B) Increases the number of WAN ports
- C) Automatically encrypts all traffic
- D) Isolates broadcast domains and allows security policies per network

58. Which protocol is commonly used to collect device monitoring information such as interface counters or CPU load?

- A) SNMP
- B) TFTP
- C) NTP
- D) SMTP

59. Which protocol keeps device clocks synchronized so that logs have consistent timestamps?

- A) NTP
- B) SNMP
- C) SSH
- D) DHCP

60. What is the benefit of DHCP reservation (address binding)?

- A) It disables DHCP on the network
- B) It encrypts DHCP messages
- C) A client always receives the same IP address based on its MAC address
- D) It increases lease time to infinite for all clients

Answer Key

#	Ans	#	Ans	#	Ans	#	Ans
1	C	16	A	31	D	46	D
2	D	17	A	32	A	47	A
3	B	18	B	33	C	48	B
4	C	19	B	34	C	49	A
5	B	20	D	35	C	50	A
6	C	21	A	36	C	51	D
7	D	22	B	37	A	52	D
8	B	23	D	38	D	53	B
9	C	24	B	39	A	54	D
10	D	25	B	40	A	55	D
11	B	26	D	41	A	56	D
12	D	27	A	42	C	57	D
13	C	28	A	43	C	58	A
14	C	29	D	44	C	59	A
15	D	30	B	45	A	60	C

Explanations

1. **C** - Routers forward packets using IP addresses at Layer 3.
2. **D** - Switches build a MAC address table and forward frames by destination MAC.
3. **B** - /26 leaves 6 host bits: $2^6 - 2 = 62$.
4. **C** - Private ranges: 10.0.0.0/8, 172.16.0.0/12 (172.16-172.31), 192.168.0.0/16.
5. **B** - Remembered as DORA.
6. **C** - ARP maps IPv4 addresses to MAC addresses within a broadcast domain.
7. **D** - UDP has no handshake or retransmission; it is used for DNS queries, VoIP, streaming.
8. **B** - NAT lets many private hosts share one or few public addresses.
9. **C** - /26 blocks are 64 addresses: 0, 64, 128, 192. 77 falls into the 64-127 block.
10. **D** - Ping sends ICMP Echo Request and receives Echo Reply.
11. **B** - DNS uses port 53 (UDP and TCP). 67 is DHCP server, 80 HTTP, 443 HTTPS.
12. **D** - Traffic outside the local subnet is sent to the default gateway.
13. **C** - 802.1Q inserts a 4-byte VLAN tag into the Ethernet frame.
14. **C** - IDs 0 and 4095 are reserved.
15. **D** - Trunk ports carry tagged frames of several VLANs; access ports belong to one VLAN.
16. **A** - STP blocks redundant paths to avoid broadcast storms.
17. **A** - 802.1w converges much faster than the original 802.1D STP.
18. **B** - LAG/LACP bundles ports; load is balanced per flow.
19. **B** - 802.3af = 15.4 W, 802.3at = 30 W, 802.3bt = 60/90 W.
20. **D** - Used with a sniffer or IDS connected to the destination port.
21. **A** - Without it, multicast is flooded to all ports in the VLAN.
22. **B** - Storm Control sets thresholds to protect the network from floods.
23. **D** - Untagged frames received on a trunk are assigned to the native VLAN (PVID).
24. **B** - The port is blocked or shut down to prevent a broadcast storm.
25. **B** - Channels are spaced 5 MHz apart but 20 MHz wide, so 1/6/11 do not overlap.
26. **D** - 802.11ax = Wi-Fi 6 (and Wi-Fi 6E in the 6 GHz band). 802.11ac = Wi-Fi 5, 802.11n = Wi-Fi 4.
27. **A** - Wi-Fi 6E extends Wi-Fi 6 into the 6 GHz band.
28. **A** - OFDMA is a key Wi-Fi 6 feature that improves efficiency, especially for many small packets.
29. **D** - 802.11ac operates only on 5 GHz; dual-band Wi-Fi 5 APs use 802.11n on 2.4 GHz.
30. **B** - Lower frequencies travel farther but offer fewer channels and more interference.
31. **D** - SAE provides forward secrecy and protects against offline brute force.
32. **A** - 5 GHz is typically less congested and faster.
33. **C** - Multi-User MIMO serves several clients at the same time.
34. **C** - 802.11r reduces handoff time; 802.11k/v assist with neighbor reports and steering.
35. **C** - Wider channels trade capacity per client against channel reuse.
36. **C** - Mesh APs relay traffic wirelessly, reducing cabling but also available bandwidth.
37. **A** - Omada integrates network devices under a central management platform.
38. **D** - The types are Hardware, Software and Cloud-Based controllers.
39. **A** - 8088 is HTTP management, 8043 is HTTPS management, 8843 is the portal HTTPS port.
40. **A** - 29810/UDP discovery; 29811-29814/TCP are used for manage, adopt, upgrade and manage-v2.
41. **A** - Sites allow managing multiple locations from one controller.
42. **C** - Devices appear as Pending until adopted.
43. **C** - Standalone is suitable for small deployments with a few devices.

- 44. **C** - Devices connect to the cloud, so remote management over the internet is possible.
- 45. **A** - ER = gateways, TL-SG = switches, EAP = access points.
- 46. **D** - Vouchers can be printed or distributed to guests with limits on duration and traffic.
- 47. **A** - Typically used in hotels, cafes and guest networks.
- 48. **B** - 802.1X authenticates devices or users before access to the LAN or WLAN is granted.
- 49. **A** - Load balancing distributes traffic; link backup is used for failover.
- 50. **A** - Link Backup provides failover, not simultaneous load sharing.
- 51. **D** - Rules match source/destination IPs or services and select an outbound interface.
- 52. **D** - Site-to-Site tunnels link entire networks; Client-to-Site is for individual remote users.
- 53. **B** - IPsec provides authentication and encryption at the network layer.
- 54. **D** - Gateway ACLs filter traffic for example between VLANs or to the WAN.
- 55. **D** - A DMZ limits exposure of the internal network if a public server is compromised.
- 56. **D** - Used to publish internal services such as a web server.
- 57. **D** - Segmentation reduces attack surface and broadcast traffic.
- 58. **A** - SNMP is used by network management systems for monitoring.
- 59. **A** - NTP synchronizes time with a server.
- 60. **C** - Useful for printers, servers and cameras.